Nomadic-Node Mobility and Ground-Robot Connectivity Across Terrains: A Protocol-in-the-Loop 5G Emulation Study for Search-and-Rescue

Innocent N. Makhubela
Dept. of Electrical Engineering
University of Cape Town
Cape Town, South Africa

Joyce Mwangama
Dept. of Electrical Engineering
University of Cape Town
Cape Town, South Africa

Author Three
Department TODO
Institution TODO
City, Country

Author Four
Department TODO
Institution TODO
City, Country

Author Five
Department TODO
Institution TODO
City, Country

Author Six
Department TODO
Institution TODO
City, Country

***Abstract*—A UAV-mounted 5G base station—a *nomadic node*—can restore on-demand connectivity to ground robots performing search-and-rescue (SAR) where terrestrial infrastructure is absent. A basic operational question is how the robots' links respond when the drone *moves*, and how that answer changes with the surrounding terrain, whose obstacles reshape the air-to-ground (A2G) channel. We answer this empirically with a protocol-in-the-loop emulation that couples a physics simulator to a real 5G standalone stack: the three-dimensional motion of a UAV-gNB and two ground-robot user equipments (UEs) continuously drives a terrain-parameterized A2G path-loss model whose output is injected at runtime into an OpenAirInterface gNB and UEs, over which genuine TCP, fixed-rate UDP video, and command-and-control (C2) traffic are measured in both directions across the emulated air interface. We instantiate five terrains—open field, urban, forest, hills, and an enterable building—each sharing one obstacle geometry between the rendered world and the radio channel, and run each as a 6-hour trial split into a *static*-gNB phase and a *nomadic*-gNB phase while both robots patrol. The effect of drone mobility on connectivity is strongly terrain-dependent and changes sign with how pervasively the terrain blocks the link: nomadism rescues a weak-link robot in line-of-sight (open field, C2 availability $56.5\% \rightarrow 88.7\%$), is neutral under localized building shadowing (urban, $\sim 96\%$ throughout), and degrades connectivity as blockage becomes pervasive (hills -10.6, forest -22 percentage points, with forest video loss exceeding 70%). We further show that maintaining connectivity in non-line-of-sight (NLoS) requires a coverage governor that reacts to *measured* path loss rather than geometric distance, and that deep outdoor-to-indoor penetration is effectively uncoverable at this link budget. We give the complete stack, the geometry-to-signal pipeline, and the emulation correctness safeguards, and release the testbed and dataset.**

***Index Terms*—UAV-enabled networks, non-terrestrial networks, nomadic base station, 5G NR, air-to-ground channel, propagation, search and rescue, protocol-in-the-loop emulation, OpenAirInterface.**

I. Introduction

Search-and-rescue (SAR) operations frequently occur where terrestrial cellular infrastructure is damaged, congested, or absent: collapsed structures, wildfire fronts, flood zones, remote wilderness. A UAV carrying a base station—a *nomadic node*—can restore connectivity on demand, serving ground robots that stream sensor data and live video to a command post while receiving navigation waypoints and safety-critical commands in return [1]–[3]. The concept and its analytic A2G channel models are well established [4], [5]. Because the base station is itself mobile, a mission planner faces a decision a fixed tower never poses: as the robots spread through the search area, should the drone *hover* at a vantage point or *roam* to follow them?

The answer cannot be geometry alone, because the robots operate *in terrain*. Buildings cast radio shadows; tree canopies attenuate in proportion to the foliage path length; hills diffract; walls impose outdoor-to-indoor penetration loss. Each mechanism loads the A2G channel differently, and moving the drone changes the *propagation path* through that terrain—sometimes clearing an obstacle, sometimes introducing one. What is needed is *system-level* evidence: how does a complete 5G protocol stack—with its scheduler, link adaptation, hybrid ARQ, and radio-link-failure (RLF) machinery—report the connectivity of two ground UEs as the aerial cell moves, and how does that report differ from one terrain to the next?

This paper answers that question empirically. We build a protocol-in-the-loop emulation in which physics-simulated UAV and robot motion continuously modulates a live 5G standalone stack carrying measured bidirectional traffic, and we instantiate five terrains that share a single obstacle geometry between the rendered world and the injected channel. Our contributions are:

- **A mobility- and terrain-coupled emulation architecture** (Section IV–V): the full software stack, its inter-component communication, the geometry-to-signal pipeline, and a terrain-parameterized A2G channel whose NLoS excess is evaluated on the same geometry the physics world renders.
- **A controlled static-versus-nomadic study across five terrains** (Section IX): each a 6-hour trial reporting both UEs' throughput, latency, coverage, and outage, establishing that the sign of the mobility effect flips with the pervasiveness of terrain blockage.
- **A propagation-aware coverage governor** (Section VII) that keeps robots connected in NLoS by reacting to measured path loss, without which cluttered terrains suffer an unrecoverable RLF cascade, together with the emulation correctness safeguards that make the measurements honest.

II. Related Work

Al-Hourani *et al.* [4] derived the widely used LoS-probability model for low-altitude platforms and showed that an optimal altitude balances free-space loss against shadowing; our channel core adopts their formulation. Surveys by Zeng *et al.* [1], [2] and Mozaffari *et al.* [3] catalogue UAV-network design opportunities, and 3GPP studied aerial channel characteristics and outdoor-to-indoor penetration for cellular UAV support [5], [6]. Terrain-specific excess loss has long-standing empirical models: the Weissberger vegetation model for foliage [7] and knife-edge diffraction for terrain obstruction [8]. This body of work is predominantly analytic or link-level. We complement it with protocol-stack-in-the-loop measurement over real bidirectional traffic, and—in contrast to the common open-field or single-environment setting—*across five distinct terrains under a common testbed*. On tooling, OpenAirInterface (OAI) provides an open, 3GPP-compliant RAN with a runtime channel-model interface [9]; Open5GS provides the 5G core [10]; Gazebo provides robotics physics [11]. To our knowledge the specific combination—physics-simulated multi-robot mobility continuously modulating a live 5G SA stack that carries measured bidirectional traffic, parameterized across terrains—has not previously been characterized for SAR.

III. Why Protocol-in-the-Loop Emulation

Three methods exist for studying A2G links: closed-form analysis, packet-level simulation (e.g., ns-3), and protocol-in-the-loop emulation with a real stack. Analysis and simulation are fast and scalable but abstract away the very mechanisms whose behaviour we wish to observe—the MAC scheduler's reaction to changing SINR, HARQ retransmission under fading, modulation-and-coding adaptation, and the RLF/re-establishment state machine. Because our question is precisely how these protocol behaviours manifest as user-visible throughput, latency, and outage as the drone moves through terrain, we run the *actual* 5G software and let those behaviours emerge, injecting only the physical-layer channel. This trades scale (one cell, two UEs) for fidelity—an appropriate trade for a study whose claims are about mechanism and operational rules, not capacity planning.

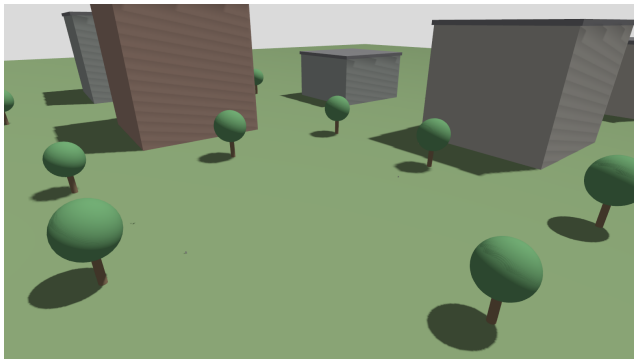


Fig. 1. A rendered terrain instance (urban). The x500 UAV carries the gNB; two TurtleBot3 robots are the ground UEs. The same building footprints seen here are imported by the channel bridge so the radio shadowing matches the visible scene.

TABLE I
Software stack and roles

Component	Version	Role
Ubuntu Linux	22.04 LTS	Host OS (single machine)
Gazebo Sim (Harmonic)	8.x	Physics; publishes 3-D poses
gz-transport / gz-msgs	13 / 10	Pub/sub bus (Python)
PX4 Gazebo model	x500	UAV/gNB carrier
TurtleBot3 (Gazebo)	burger	Two ground-robot UEs
OpenAirInterface	2026.w2x	5G NR SA RAN + rfsimulator
Open5GS	2.8	5G core (AMF/SMF/UPF/…)
MongoDB	6.0	Subscriber database
Python	3.10	Bridge, mobility, analysis
iproute2	5.15	Per-robot network namespaces
iperf3	3.16	TCP/UDP generation & measurement

IV. System Architecture

The testbed runs three concurrent control loops on a single host (Fig. 2); Fig. 1 shows a rendered instance. The *channel loop* (2 Hz) reads poses, computes terrain-aware A2G path loss, and injects it into the live RAN. The *traffic loop* (∼one cycle per robot every 15–20 s) exercises and measures each link. The *mobility loop* flies the UAV and drives the robots. Table I lists the inventory; we motivate each choice below.

Physics and terrain (Gazebo). Gazebo provides rigid-body motion, a transport bus for streaming ground-truth poses at ∼10 Hz, and standard robot models (PX4 x500 airframe carrying the gNB; TurtleBot3 ground robots as UEs). Each terrain is a Gazebo world whose obstacle geometry—building footprints, tree positions, terrain height field, or building walls—is authored once and *shared* with the channel model, so the radio shadowing is consistent with the visible scene by construction.

5G RAN (OpenAirInterface). OAI is a real, 3GPP-compliant 5G NR standalone implementation, so the scheduler, HARQ, link adaptation, and RLF are genuine, not modelled. Its `rfsimulator` backend replaces the RF front-end with a software channel between gNB and UE I/Q

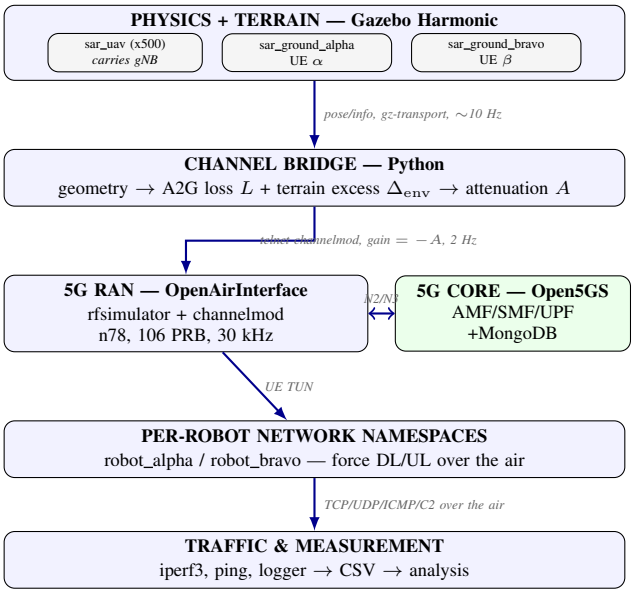


Fig. 2. Testbed architecture. The channel loop (pose → bridge → RAN) closes the coupling between physical motion *in terrain* and radio conditions; user traffic is measured over the emulated air interface after namespace isolation.

streams and exposes a runtime channel-model control interface (`channelmod`) through which per-link path loss can be updated live—exactly the hook the mobility coupling requires. srsRAN was considered but its channel emulation is less amenable to live per-link injection; commercial stacks are closed; pure simulators lack the real protocol implementation that is the point of the study.

5G core (Open5GS) and subscribers. Open5GS supplies a production-grade 5G core (AMF/SMF/UPF and supporting functions) with a MongoDB subscriber database, so UE attach and PDU-session establishment are genuine NAS/NGAP/PFCP procedures rather than shortcuts. Each UE terminates in a host TUN interface.

Traffic-path isolation (namespaces). Linux network namespaces move each UE's TUN into its own namespace after attach; the modem keeps its open file descriptor across the move, so the RAN is untouched, but base-to-robot traffic now genuinely traverses core→gNB→radio in both directions (Section VII-A). `iperf3` generates and measures flows; `ping` measures latency and delivery; a logger joins each cycle's measurements with the concurrent geometry into one CSV row.

V. Geometry and the Terrain-Parameterized A2G Channel

The channel is driven entirely by instantaneous geometry. With the UAV at (x_u, y_u, h_u) and a robot at (x_r, y_r, h_r), the horizontal distance, slant distance, and elevation angle are

$$d_h = \sqrt{(x_u - x_r)^2 + (y_u - y_r)^2}, \tag{1}$$

$$d_{3D} = \sqrt{d_h^2 + (h_u - h_r)^2}, \tag{2}$$

$$\theta = \arctan\big[(h_u - h_r)/d_h\big] \ (\text{deg}). \tag{3}$$

Equations (1)–(3) encode the core insight: two robots at equal map distance d_h can experience very different links, because the radio responds to the slant path and, decisively, the elevation angle. The free-space loss over the slant path at carrier frequency f is

$$L_{\text{FSPL}} = 20\log_{10}(d_{3D}) + 20\log_{10}(f) - 147.55. \tag{4}$$

Following [4], the LoS probability is a sigmoid in elevation angle and the base loss adds an angle-dependent excess:

$$P_{\text{LoS}}(\theta) = \frac{1}{1 + a\,\exp[-b(\theta - a)]}, \tag{5}$$

$$L_0 = L_{\text{FSPL}} + P_{\text{LoS}}\,\eta_{\text{LoS}} + (1 - P_{\text{LoS}})\,\eta_{\text{NLoS}}, \tag{6}$$

with $a = 9.61$, $b = 0.16$, $\eta_{\text{LoS}} = 1$ dB, $\eta_{\text{NLoS}} = 20$ dB.

Terrain excess. The distinguishing element of this study is an additive terrain term Δ_{env} evaluated on the *same* geometry that Gazebo renders (Table II). For *urban*, if the UAV→robot segment intersects a building footprint the link is hard-NLoS (excess up to η_{NLoS}); otherwise it is clear. For *forest*, Δ_{env} grows with the foliage path length through tree canopies (Weissberger form [7]). For *hills*, it is knife-edge diffraction over the terrain profile [8]. For *indoor*, a 3GPP outdoor-to-indoor term applies whenever the robot is inside the building footprint—a fixed wall-penetration loss plus a per-metre depth loss [6]. The total loss is

$$L = L_0 + \Delta_{\text{env}}(\text{UAV}, \text{robot}, \text{geometry}) + X_\sigma, \tag{7}$$

where X_σ is correlated log-normal shadow fading (larger variance under NLoS), decorrelated over distance moved so a stationary robot holds a steady offset. The bridge references L to $L_{\text{ref}} = L_0(d{=}50\,\text{m}, h{=}30\,\text{m})$, offsets by a per-link baseline L_{base} (8 dB for α's uplink, 13 dB for β's—a 5 dB gap emulating a weaker device class), and clips:

$$A = \text{clip}\big(L_{\text{base}} + (L - L_{\text{ref}}),\ 0,\ 45\big)\ \text{dB}. \tag{8}$$

Because OAI's channel field is applied as a linear *gain* (Section VII-A), the value written to the RAN is $-A$, scaling each I/Q sample by $10^{-A/20}$—a true attenuation. Fig. 3 shows the emulated result matches the model to ∼1.2 dB.

TABLE II
The five terrains and their NLoS mechanism

Terrain	Obstacle	Excess Δ_{env}
Open field	none	graded angle excess only (LoS)
Urban	building footprints	hard blockage (segment test)
Forest	tree canopies	foliage path length (Weissberger)
Hills	terrain height field	knife-edge diffraction
Indoor	enterable building	3GPP O2I: wall + depth loss

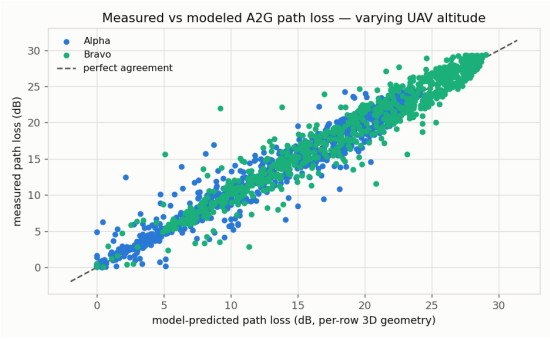


Fig. 3. Channel validation: emulated injected path loss vs. per-sample model prediction, on the diagonal (MAE ∼1.2 dB). The measurements below are a consequence of the intended physics, not artifacts.

VI. Traffic Model and Metrics

Each measurement cycle exercises a UE with three traffic types mirroring SAR usage. **Bulk upload (TCP):** a short `iperf3` transfer (robot→base, slow-start excluded), modelling a sensor dump; goodput reflects reliable-transport capacity. **Video (UDP):** a fixed 8 Mbit/s uplink stream modelling a live camera; delivered rate, loss, and jitter reflect stream quality. **Command & Control (C2):** five 64-byte ICMP pings *base→robot*, emulating the downlink control channel (navigation waypoints, emergency-stop). C2 is the safety channel: its RTT and loss measure *controllability*—the ability to steer or halt a robot—not bandwidth. We also record UE-reported RSRP and SINR, and define *coverage/C2 availability* (fraction of samples with less than 100% C2 loss), *outage probability* (fraction in total loss), and inter-robot Jain fairness. Because both robots are logged every cycle, all metrics are reported per UE (α = stronger budget, β = weaker).

VII. Mobility and Propagation-Aware Coverage Governance

Robots drive by differential-drive velocity commands at 1 m/s along randomized continuous headings—genuine kinematics, not teleportation. In each trial the UAV first *hovers* statically, then performs a continuous three-dimensional random walk over a bounded box (Section VIII). To keep robots from being stranded as coverage moves, a governor confines each robot. A naive governor uses a geometric *safe radius* from the open-field model,

$$R_{\text{safe}}(h) = \max\{d_h : L_{\text{base},\beta} + [L_0(d_h, h) - L_{\text{ref}}] \le A_{\text{safe}}\}, \tag{9}$$

with A_{safe}=30 dB (≈5 dB below the RLF cliff), evaluating to 101/131/163 m at h=20/40/60 m. **This fails in NLoS:** a robot can sit well inside R_{safe} yet be blocked by a building or canopy (path loss near the cliff), because the distance rule is blind to obstacles. Worse, once such a UE loses the link it declares RLF, and because a UE cannot complete cell (re)synchronization while buried in deep NLoS, restarting it *fails to re-attach*—the recovery loops indefinitely and the robot stays dark. We therefore make the governor react to *measured* path loss: when a UE's loss is elevated it is steered toward the gNB ground track (raising the elevation angle and reducing NLoS depth), and if loss stays near the cliff the robot is relocated to a small radius from the gNB *before* it dies. For the indoor mission, where the robot must occupy the building, it instead retreats toward the door (reducing penetration depth) and re-enters on recovery. This prevention-first design converts an unrecoverable cascade into rare, short outages (Section IX).

A. Emulation correctness safeguards

Building an honest single-host testbed surfaced three failure modes producing *plausible but wrong* data; all reported results were collected only after all three were fixed. *(i) The channel field is a gain, not a loss.* OAI's `channelmod` applies `path_loss_dB` as a linear gain ($10^{x/20}$)—positive values *amplify*—so positive "loss" produced a link that degraded only via sample clipping and UE RSRP that *rose* with distance; we keep A positive and negate once at injection (9). *(ii) Single-host traffic bypasses the radio.* With host-local UE IPs, base-to-robot traffic went over loopback (a "downlink" test read Gbit/s while the gNB scheduled 0.15 Mbit/s); moving each TUN into a per-robot namespace forces the air path. *(iii) A single-client control port drops writes.* OAI's telnet accepts one client per port; a second socket to the gNB port sat unaccepted in the backlog, silently discarding one UE's uplink modulation. A shared connection per port with a start-up read-back refuses to run unless every baseline verifies.

VIII. Experimental Setup

Each terrain runs a single 6-hour trial in two phases: *phase 1* (0–3 h) with the gNB hovering statically, and *phase 2* (3–6 h) with the gNB performing a continuous 3-D random walk over a bounded box (never dwelling). Both robots free-patrol throughout, so the only variable between phases is gNB mobility. The gNB uses band n78 (3.32 GHz, 40 MHz, 30 kHz SCS, 106 PRB) on Open5GS. A single physical stack must serve only one trial—two concurrent trials share the same robots and gNB and corrupt each other—so trials are serialized under an exclusive lock. Reported worlds are clean 6-hour trials with 733, 721, 482, and 974 samples for open field, urban, forest, and hills; the indoor trial is reported separately as a coverage limit. Exploratory runs collected while the safeguards of Section VII-A were latent are excluded from every number.

IX. Results

A. Propagation and coverage across terrains

Fig. 4 bins measured path loss against gNB–robot slant distance per terrain—the physical basis of everything that follows: open field and hills sit low (shallow, LoS-dominated); forest climbs steeply because a longer slant path cuts through more canopy; urban is intermediate with building-driven jumps. Fig. 5 makes the same physics spatial—the link attenuation over the search area for a static drone (left), and the fraction of time each point is covered as the drone roams (right). The terrain signature is unmistakable: open field is a clean radial field, urban is fractured by building shadows,

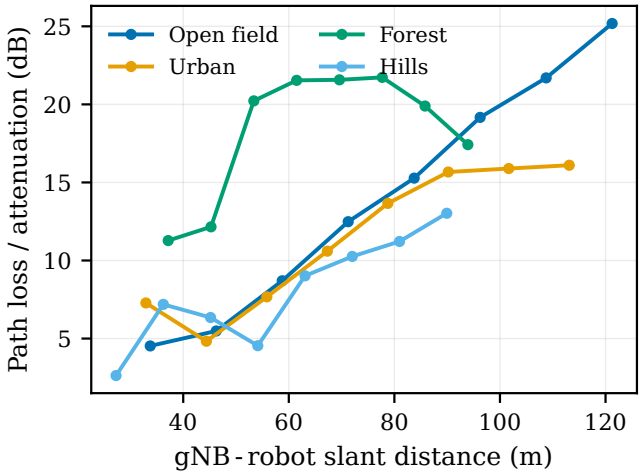


Fig. 4. Measured path loss vs. gNB–robot slant distance (binned means) per terrain. Forest rises steeply with distance; open field and hills stay low; urban is intermediate.

forest disintegrates into thin coverage fingers, and hills form an irregular covered island. Terrain does not merely shift coverage—it changes its *shape*, so moving the drone has a terrain-specific effect.

B. Watching the drone move

Fig. 6 shows the raw phenomenon: both UEs' C2 connectivity over the full 6-hour mission, per terrain, with the static→nomadic transition marked at 3 h (shaded = nomadic). The effect is legible directly. In *open field*, the weak UE β repeatedly drops while the gNB is static, then *steadies* once it roams. In *urban* both UEs stay flat across the transition. In *forest*, both are healthy while static, then connectivity *collapses* the moment the gNB begins roaming. *Hills* is a milder version. The bars below quantify it.

C. Moving the drone: the effect flips with terrain

Fig. 7 and Table III give the central result: **the effect of drone mobility on connectivity is terrain-dependent and changes sign with the pervasiveness of blockage.** In *open field* (line-of-sight) mobility *helps* the weak-link robot: β's C2 availability rises from 56.5% (static) to 88.7% (nomadic), a +32 pp gain, as the wandering gNB periodically drifts closer and rescues a robot stranded near the coverage edge (its mean distance falls from 91 to 75 m), while α stays near 98%. In *urban* mobility is *neutral* (∼96–98%): localized buildings leave LoS pockets the governor keeps robots in. In *hills* and *forest* mobility *hurts*, scaling with how pervasive the blockage is—−10.6 pp for knife-edge terrain and −22 pp for everywhere-present foliage, which collapses both UEs to ∼73–75%.

D. What it costs the mission: streaming and control

Connectivity translates directly into mission capability (Fig. 8). The 8 Mbit/s video feed—the operator's eyes—arrives nearly intact in open field, urban, and hills, but in forest a

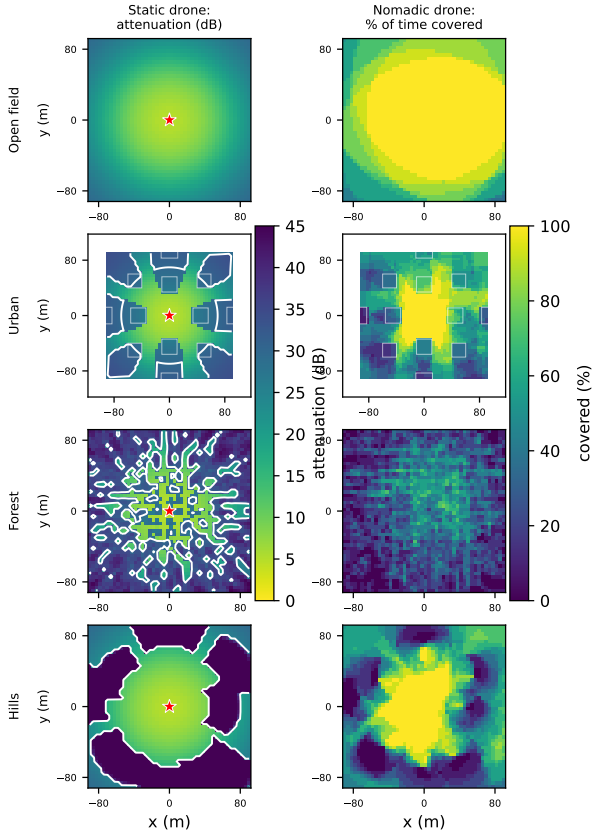


Fig. 5. Coverage field over the search area (weaker UE), per terrain. Left: link attenuation for a *static* drone at the origin (red star; white contour = usable edge). Right: fraction of time each point is covered as a *nomadic* drone roams. Urban is fractured by building shadows; forest never coheres; open field's field is radial and spreads as the drone moves.

nomadic drone drives it to 1.9 Mbit/s with over 70% loss: the operator goes blind exactly when the drone moves. Command latency, by contrast, holds near 37 ms median *whenever the link is up*, in every terrain and phase—so controllability fails as an *availability* problem (the command channel is present or absent), not a latency creep.

E. Throughput and coverage as the UE moves

Fig. 9 plots TCP goodput against UE–gNB distance, static vs. nomadic, per terrain. In open field the static curve falls off a coverage cliff with range while the nomadic drone flattens it by following the robot; in forest both curves sit low and nomadism cannot lift them. This is the mobility trade in one view: moving the drone buys range where the path is clear and buys nothing where it is blocked.

F. The real stack adapts

Because we run a genuine RAN, link adaptation is observed, not modelled (Fig. 10). As the forest-nomadic channel de-

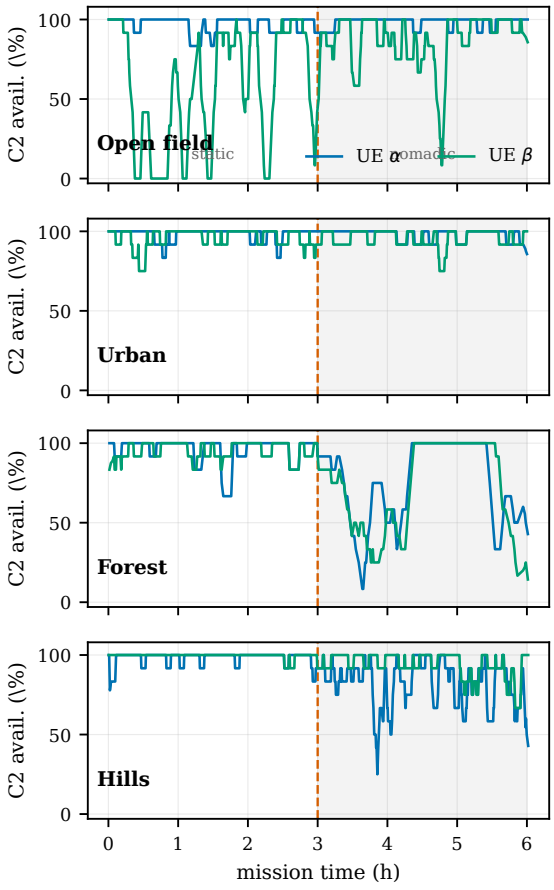


Fig. 6. C2 connectivity over the 6-hour mission, per terrain, both UEs (rolling availability); dashed line = static→nomadic transition (3 h), shaded = nomadic. UE β steadies under nomadism in open field; forest collapses when the drone starts moving.

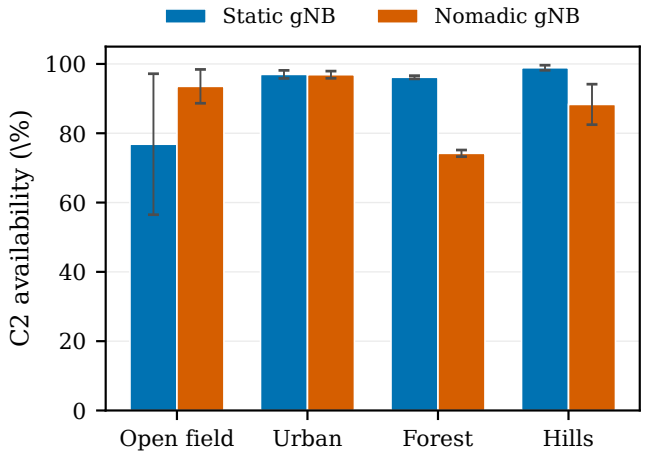


Fig. 7. C2 availability by terrain and gNB phase (bar = robot mean; whiskers span the two UEs). Nomadism helps in open field, is neutral in urban, and hurts in hills and forest.

TABLE III
C2 AVAILABILITY AND MEAN LINK METRICS BY TERRAIN AND GNB PHASE (ROBOT-AVERAGED). Δ IS NOMADIC−STATIC C2 (PP).

Terrain	Phase	C2 (%)	PLoss (dB)	SINR (dB)	TCP (Mb/s)
Open field	Static	76.8	16.5	35.1	9.7
	Nomadic	93.5	14.2	38.8	13.1
	Δ	**+16.7**			
Urban	Static	97.0	8.8	43.8	17.6
	Nomadic	96.9	10.4	42.6	16.1
	Δ	**−0.1**			
Hills	Static	98.9	3.9	45.7	21.6
	Nomadic	88.3	9.3	44.2	19.4
	Δ	**−10.6**			
Forest	Static	96.2	10.8	40.9	15.1
	Nomadic	74.2	21.7	32.5	3.2
	Δ	**−22.0**			

grades, the scheduler backs uplink spectral efficiency down ($4.2 \rightarrow 2.7$ bits/sym) while HARQ holds block error near target ($< 0.04\%$)—the protocol trading rate for reliability, which a link-level abstraction would miss. This is the payoff of protocol-in-the-loop emulation.

G. Device budget and the weakest radio

Consistent with the open-field baseline, the two UEs differ only by 5 dB of baseline budget yet experience qualitatively different networks: α is distance-limited while β is budget-limited, reaching the cliff sooner and defining the team's usable region. High Jain fairness ($J \approx 0.97$) confirms the disparity is device budget, not scheduler bias—so the remedy is coverage margin, and the governor and drone placement must be sized for the *weakest* radio, in every terrain.

H. Keeping robots connected in NLoS

Fig. 11 counts link-recovery events per 6-hour trial. With the propagation-aware governor, the RLF cascade is eliminated in urban and hills (zero failed re-attaches) and contained in forest, where the governor intervenes 574 times—roughly once every 37 s—to keep robots out of foliage dead zones. The cost of prevention is a more gNB-centred robot distribution: the governor keeps NLoS robots closer to the gNB than free roaming would (the concentrated nomadic coverage of Fig. 5). This is the price of any valid measurement in NLoS; without it the cluttered terrains do not merely degrade, they fail to produce data.

I. Indoor: a coverage limit

The outdoor-to-indoor terrain is qualitatively different (Fig. 12). The fixed wall-penetration term plus the link baseline places the *entire* building interior near the RLF cliff—depth barely matters—so shadow spikes trip RLF repeatedly, and each ∼2-minute re-attach dominates the timeline. Lowering the gNB to hover directly overhead (cutting ∼7 dB of A2G loss) and adding the retreat-to-door recovery removes the cascade and keeps robots alive, but C2 availability remains only 12–18%. We read this as a genuine finding: a ground robot cannot hold connectivity deep inside a building at this link budget, and *leaving the building is the only recovery*.

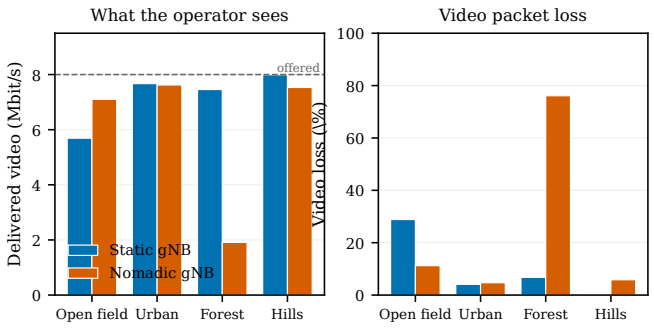


Fig. 8. Delivered video rate and loss, static vs. nomadic gNB per terrain. Forest video collapses to 1.9 Mbit/s ($> 70\%$ loss) under a nomadic drone; open-field nomadism instead *helps* the weak UE.

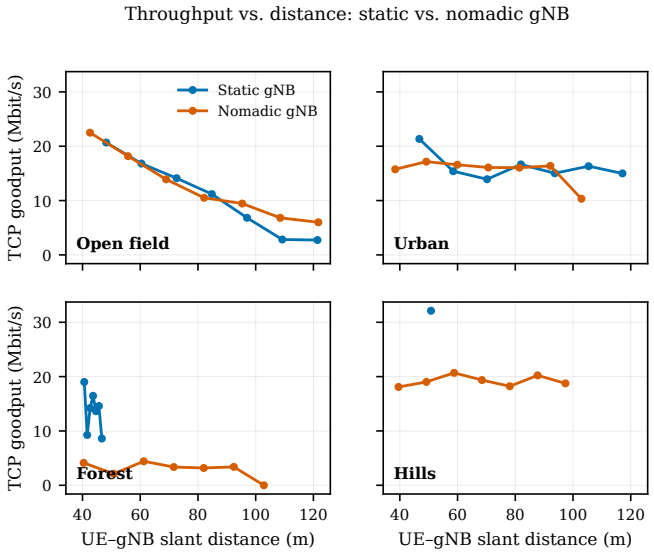


Fig. 9. TCP goodput vs. UE–gNB distance, static vs. nomadic gNB, per terrain. Nomadism flattens the open-field cliff (the drone follows the robot) but cannot lift the blocked forest link.

X. Discussion

The unifying explanation is that moving the drone changes the *propagation path*, and whether that helps depends on how blockage is distributed. In LoS a moving gNB is pure geometric benefit—it can shorten links and rescue edge robots—so nomadism helps the weak robot most. Where blockage is *localized* (urban buildings), a governor can keep robots in LoS pockets and mobility washes out. Where blockage is *pervasive* (forest canopy) there is no pocket to hold: any repositioning that lengthens the ground path lengthens the foliage path, so mobility is strictly harmful, and the damage scales with blockage density (hills $<$ forest). Two operating rules follow for SAR: *hover* a nomadic node when the environment is cluttered and pervasive, and *roam* it only when links are LoS-dominated or robots risk stranding at the coverage edge; and size coverage for the weakest radio in the team.

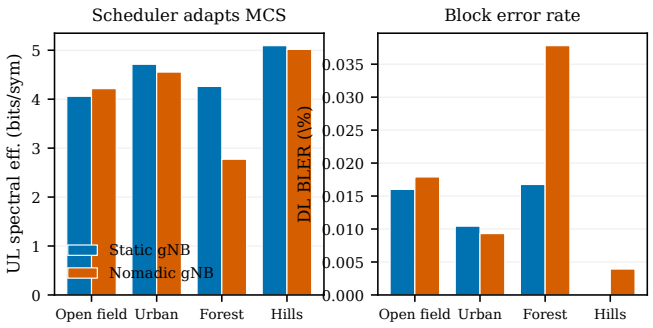


Fig. 10. Uplink spectral efficiency and downlink BLER, static vs. nomadic per terrain. In forest-nomadic the scheduler drops MCS while HARQ keeps BLER bounded—the real stack adapting.

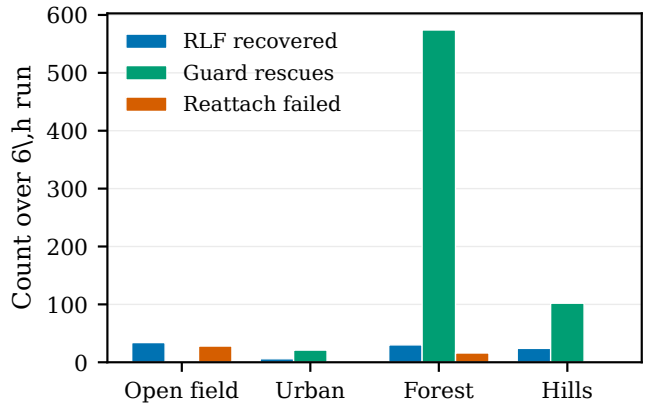


Fig. 11. Link-recovery events per 6-hour trial. Forest requires by far the most governor interventions; urban and hills incur zero failed re-attaches.

XI. Limitations

The RAN runs over `rfsimulator`, so absolute link budgets depend on model constants and the configured noise floor; we therefore emphasize *relative* static-vs-nomadic effects within a fixed calibration rather than absolute edges. The coverage governor is necessary to obtain data in NLoS but alters robot trajectories (more gNB-centred), so the reported effects are conditional on prevention-based coverage keeping; free-roaming robots would simply lose the link. We study one cell and two UEs; multi-cell handover and many-robot contention are future work. Under outage the per-sample probes time out, so heavily degraded phases yield fewer samples (e.g., forest phase 2), though fine-grained path loss is logged continuously by the bridge. The indoor result reflects one building geometry and link budget. Subject to these, the *direction* of the findings follows from geometry, link budget, and protocol-state mechanisms that carry to physical LoS/NLoS deployments.

XII. Conclusion and Future Work

We presented a mobility- and terrain-coupled 5G emulation testbed for UAV-mounted nomadic nodes serving ground robots, and used it to measure how two ground UEs report

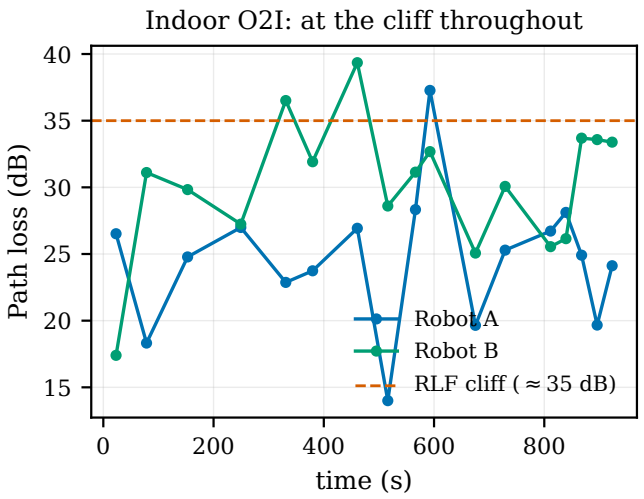


Fig. 12. Indoor O2I path loss over time (low-hovering gNB). Both robots sit at the RLF cliff throughout; the interior offers no reliably covered pocket.

their connectivity as the drone moves, across five terrains. The effect of gNB mobility on connectivity is terrain-dependent and changes sign with the pervasiveness of blockage: helpful in line-of-sight (open field, $+32$ pp for the weak robot), neutral under localized urban shadowing, and increasingly harmful under diffraction and foliage (hills -10.6, forest -22 pp), while deep outdoor-to-indoor penetration is effectively uncoverable. We further showed that maintaining connectivity in NLoS requires a governor that reacts to measured path loss rather than geometric distance, and we documented the emulation safeguards that make single-host measurement honest. Future work relaxes the single-cell assumption toward multi-node handover, adds a coverage-edge recovery-time study, and pursues SDR hardware validation.

Acknowledgment

[Anonymized for double-blind review.]

References

[1] Y. Zeng, R. Zhang, and T. J. Lim, "Wireless communications with unmanned aerial vehicles: Opportunities and challenges," *IEEE Commun. Mag.*, vol. 54, no. 5, pp. 36–42, 2016.

[2] Y. Zeng, Q. Wu, and R. Zhang, "Accessing from the sky: A tutorial on UAV communications for 5G and beyond," *Proc. IEEE*, vol. 107, no. 12, pp. 2327–2375, 2019.

[3] M. Mozaffari, W. Saad, M. Bennis, Y.-H. Nam, and M. Debbah, "A tutorial on UAVs for wireless networks: Applications, challenges, and open problems," *IEEE Commun. Surveys Tuts.*, vol. 21, no. 3, pp. 2334–2360, 2019.

[4] A. Al-Hourani, S. Kandeepan, and S. Lardner, "Optimal LAP altitude for maximum coverage," *IEEE Wireless Commun. Lett.*, vol. 3, no. 6, pp. 569–572, 2014.

[5] 3GPP, "Study on enhanced LTE support for aerial vehicles," TR 36.777, 2017.

[6] 3GPP, "Study on channel model for frequencies from 0.5 to 100 GHz," TR 38.901, 2020.

[7] M. A. Weissberger, "An initial critical summary of models for predicting the attenuation of radio waves by trees," ECAC-TR-81-101, 1982.

[8] ITU-R, "Propagation by diffraction," Recommendation P.526.

[9] N. Nikaein *et al.*, "OpenAirInterface: A flexible platform for 5G research," *ACM SIGCOMM Comput. Commun. Rev.*, vol. 44, no. 5, pp. 33–38, 2014.

[10] Open5GS, "Open source implementation of the 5G core network," https://open5gs.org, accessed 2026.

[11] N. Koenig and A. Howard, "Design and use paradigms for Gazebo, an open-source multi-robot simulator," in *Proc. IEEE/RSJ IROS*, 2004, pp. 2149–2154.